

ETQ-303 v3

Exact 303-State Root-Indexed Event Protocol

Formal Mathematical Specification
and Reproducibility Record

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Archival note. This document is a documentation-only successor to the immutable ETQ-303 v3.0.0 release identified above. Version 3.0.1 clarifies algebraic dimension and tensor notation; it does not alter the mathematical construction, state indexing, event protocol, deterministic artifacts, implementation identity, or scientific claim boundary. The v3.0.0 tag and archived files remain unchanged.

Abstract

ETQ-303 v3.0.1 documents the exact finite event protocol first released as ETQ-303 v3.0.0 and constructed from the released ETQ-101 v2.0.0 selected-root model. The preserved base has 101 E8-root labels arranged as two triality-fixed singlets and 33 complete order-three D4 triality orbits. Version 3 forms the tensor state space

$$\mathcal{H}_{303} = \mathcal{H}_{101} \otimes \mathbb{C}^3,$$

so every preserved site carries an independent three-state external fibre. A declared monomial step acts on support as

$$(j, a) \mapsto (j + 1 \bmod 101, a + 1 \bmod 3)$$

and carries exact Gaussian-unit phase exponents $(3, 2, 3)$, with aligned symbols $(-i, -1, -i)$. Coprimality of 101 and 3 gives one 303-state support orbit, while the three-step phase product equals one; hence the complete monomial operator has exact order 303.

The graph context is the Cartesian product $G_{303} = G_{101} \square C_3$. It has 303 vertices, 5,364 edges, degree range 24–57, and one connected component. Here 303 is the algebraic dimension and the number of basis-indexed states in a finite complex vector space, not a number of physical spacetime dimensions. The canonical output is a receiver-neutral 303-entry event document. JSON, CSV, graph JSON, atlas JSON, and symbolic MIDI artifacts are deterministic projections; no receiver is mathematically privileged. Identity-bearing JSON uses safe integers only, recursive lexicographic key order, domain-separated SHA-256 receipts, and an implementation source-bundle identity. The root workflow emits no hertz, sample rate, waveform, PCM, WAV, floating eigensolver identity, or empirical validation claim.

Contents

1	Purpose, scope, and normative status	1
1.1	Version lineage and archival identity	1
1.2	Notation and Conventions	1
2	Epistemic layers and interpretation rules	1
3	Preserved ETQ-101 v2 base	2
3.1	Exact E8 root realization	2
3.2	Embedded D4 triality	2
3.3	Preserved selector and 101-state base	3
3.4	Preserved selected-root graph	3
4	The 303-state tensor extension	3
4.1	Canonical tensor index	4
5	Exact monomial step and Gaussian phases	4
6	Exact order 303	5
7	Canonical event ordering and CRT inverse	5
8	Site registry and preserved root identity	5
9	Cartesian graph lift	6
9.1	Structural spectral consequence	6
10	Canonical event document	7
11	Receiver artifacts and root output policy	7

12 Canonical serialization and domain-separated hashes	8
13 Implementation identity and acyclic provenance	8
14 Build safety	9
15 Breaking change from v2	9
16 Auxiliary Tanner and spectral modules	10
17 Reproducibility and verification	10
18 Zenodo v3.0.1 deposit package	11
19 Claim matrix and reviewer-facing limits	11
20 Conclusion	12
A Canonical parameter summary	12
B Required deterministic checks	13
C Optional extension boundaries	13

1 Purpose, scope, and normative status

This manuscript is the human-readable formal specification and reproducibility record for the ETQ-303 v3.0.1 documentation patch. It describes the exact protocol and machine-readable identity first archived as ETQ-303 v3.0.0. The current machine-readable repository fixture is `examples/etq-303.v3.canonical.json`; its schema is `spec/etq-303.v3.schema.json`. The exact construction is implemented by the dependency-free Node.js modules named in the implementation identity, and the release verification endpoint is `npm run verify`.

The supported forward claim, unchanged from v3.0.0, is:

ETQ-303 v3.0.0 is an exact finite 303-state tensor extension of the preserved ETQ-101 v2 selected-root basis. It defines a single 303-step monomial support traversal with exact Gaussian-unit phase labels, the Cartesian graph lift $G_{101} \square C_3$, a canonical event document, and deterministic allowlisted receiver artifacts with acyclic provenance receipts.

The word *exact* refers to finite integer combinatorics, modular arithmetic, and finite-dimensional linear algebra after the declared v2 selector and v3 protocol choices have been fixed. It does not mean that the selector, external fibre, starting address, receiver mapping, or serialization vocabulary is uniquely selected by E8 or by nature.

1.1 Version lineage and archival identity

Version 3 preserves the ETQ-101 v2.0.0 base release, its source tag, selected basis, adjacency, degree-potential numerators, and contract receipt [1]. The v3 implementation was merged through repository pull request 12 at commit `6b55e51647226d1c248dc8d79f9ed9336241c2ac`, frozen at tag `v3.0.0`, and archived at DOI [10.5281/zenodo.21455181](https://doi.org/10.5281/zenodo.21455181).

Version 3.0.1 is a terminology and exposition clarification only. It does not replace, retag, or silently rewrite the v3.0.0 archive. The published clarification is archived separately at DOI [10.5281/zenodo.21494678](https://doi.org/10.5281/zenodo.21494678).

1.2 Notation and Conventions

Unless explicitly stated otherwise, scalars are elements of $\mathbb{C}$, all vector spaces appearing in the construction are finite-dimensional, and $\dim(V)$ denotes the algebraic dimension of a vector space V . The symbol $\otimes$ denotes the tensor product over $\mathbb{C}$ unless another scalar field is explicitly declared. Operators act on the finite-dimensional spaces declared in the construction; the real Euclidean root-coordinate spaces used to realize root systems are identified explicitly when they occur.

Terminology (Dimensions). Throughout this document, *dimension* refers exclusively to finite-dimensional vector-space dimension, including tensor-product spaces, unless another mathematical meaning is explicitly declared. It does not refer to physical spacetime dimensions. References to the rank or dimensionality of algebraic structures such as D_4 , E_8 , root spaces, selected-state spaces, or tensor-product spaces are purely mathematical unless an additional physical interpretation is explicitly introduced. No physical interpretation is to be inferred from dimension, rank, state count, or tensor-product notation alone.

2 Epistemic layers and interpretation rules

To prevent reproducibility from being confused with mathematical necessity or empirical validation, statements in this specification belong to the following layers.

Layer	Meaning in this specification
Preserved exact base	The 240 standard E8 roots, embedded order-three D4 triality, v2 orbit decomposition, selected 101 root labels, selected adjacency, and preserved v2 receipts.
Exact v3 consequence	Tensor dimension, tensor-index bijection, CRT traversal, phase closure, order 303, Cartesian graph-product counts, degrees, and connectivity.
Authored protocol	Promotion of the optional product extension, selector-order shift R_{101} , starting address (0,0), event field names, external fibre semantics, and receiver profiles.
Symbolic receiver	JSON, CSV, graph/atlas JSON, and MIDI serializations. These are deterministic encodings, not physical spectra or unique observations.
Numerical auxiliary	Floating eigensolvers, spectral ratios, frame-bin calibration, graph surrogates, or rendered laboratory experiments. None enters v3 canonical identity.
Empirical/perceptual	Any listener study, measured system, physical qutrit, acoustic apparatus, or falsifiable physical interpretation. No such validation is claimed.

“Canonical” means canonical within the named v3 contract. A changed base selector, fibre action, event ordering, source-file identity set, or receiver mapping defines a different named profile and requires new receipts.

3 Preserved ETQ-101 v2 base

3.1 Exact E8 root realization

Use the standard 240-root coordinate realization in the eight-dimensional Euclidean root space $\mathbb{R}^8$ [2, 3, 4]:

$$\mathcal{R}_{E_8} = \{\pm e_i \pm e_j : i < j\} \cup \left\{ \frac{1}{2}(s_1, \dots, s_8) : s_i \in \{-1, 1\}, \#\{i : s_i = -1\} \text{ even} \right\}. \quad (1)$$

The first family contributes 112 roots and the second contributes 128. Every root has squared norm two. The implementation stores doubled integer coordinates $\tilde{r} = 2r$, so

$$\tilde{r} \cdot \tilde{r} = 8, \quad r_a \cdot r_b = 1 \iff \tilde{r}_a \cdot \tilde{r}_b = 4. \quad (2)$$

Root identity, sorting, triality, and adjacency therefore use exact integers. The numeral 8 here is the rank and Euclidean root-space dimension of E_8 ; it is not a spacetime dimension.

3.2 Embedded D4 triality

On each four-coordinate block define

$$A = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \end{pmatrix}, \quad \tau = A \oplus A. \quad (3)$$

Each block realizes the rank-4 D_4 root-space action. The numeral 4 in D_4 refers to algebraic rank, equivalently the dimension of its Euclidean root space, and not to four-dimensional spacetime. Direct arithmetic gives $A^\top A = I_4$ and $A^3 = I_4$. Exact enumeration verifies

$$\tau \mathcal{R}_{E_8} = \mathcal{R}_{E_8}, \quad \tau^3 = I_8. \quad (4)$$

The induced root permutation has 12 fixed roots and 76 three-cycles:

$$12 + 76 \cdot 3 = 240. \quad (5)$$

This is D4/Spin(8) triality embedded in an E8-derived construction, not an outer triality intrinsic to E8 itself [3].

3.3 Preserved selector and 101-state base

The v2 selector sorts doubled coordinate vectors lexicographically, retains the first two fixed roots, takes the lexicographically least representative of each non-fixed orbit, and retains the first 33 complete orbits. In the preserved ordering,

$$S_{101} = \{s_0, s_1\} \cup \bigcup_{m=0}^{32} \{r_m, \tau r_m, \tau^2 r_m\}. \quad (6)$$

The root coordinates label an abstract orthonormal graph basis:

$$\mathcal{H}_{101} = \ell^2(S_{101}) = \mathbb{C}^2 \oplus (\mathbb{C}^{33} \otimes \mathbb{C}^3), \quad \dim \mathcal{H}_{101} = 2 + 33 \cdot 3 = 101. \quad (7)$$

Thus the ETQ-101 selected basis contains 101 states. This state-space dimension is distinct from the eight-dimensional Euclidean coordinates used to label its selected roots. It is not a 101-dimensional representation of the E8 Lie algebra; standard representation tables provide the relevant comparison [5].

3.4 Preserved selected-root graph

For $r_a, r_b \in S_{101}$ define

$$(W_{101})_{ab} = \begin{cases} 1, & a \neq b \text{ and } r_a \cdot r_b = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

The exact selected graph G_{101} has 101 vertices, 1,687 edges, degree sum 3,374, degree range 22–55, and one connected component. The exact centered degree-potential numerators are

$$\nu_j = 101d_j - 3374, \quad \max_j |\nu_j| = 2181, \quad \sum_j \nu_j = 0. \quad (9)$$

Version 3 preserves the labelled degree sequence and these v2 receipts. It does not replace the graph with a 101-cycle or sort degrees independently of their vertex labels.

4 The 303-state tensor extension

Definition 1 (ETQ-303 state space). The v3 state space is

$$\boxed{\mathcal{H}_{303} = \mathcal{H}_{101} \otimes \mathbb{C}^3}. \quad (10)$$

Its canonical basis consists of vectors $|j, a\rangle$ with $j \in \mathbb{Z}_{101}$ indexing a preserved v2 site and $a \in \mathbb{Z}_3$ indexing an independent external fibre.

Proposition 2 (Dimension). *The algebraic dimension of the ETQ-303 state space is*

$$\dim(\mathcal{H}_{303}) = \dim(\mathcal{H}_{101}) \dim(\mathbb{C}^3) \quad (11)$$

$$= 101 \times 3 \quad (12)$$

$$= 303. \quad (13)$$

Proof. The dimension of a finite tensor product is the product of dimensions. $\square$

Thus 303 is both the dimension of the finite event/state vector space and the number of basis-indexed states in the exact construction. It is not a claim of 303 physical dimensions. The factor $\mathbb{C}^3$ is the declared independent three-state, or qutrit-like, tensor factor used by the protocol; the tensor product introduces no additional physical ontology.

The 303 vectors are root-indexed tensor states. There remain exactly 101 distinct selected E8 root labels. The external fibre is not the pre-existing internal triality label on the 99-dimensional non-fixed sector. The two fixed singlets have no internal triality label, but each fixed singlet has all three external fibre states.

4.1 Canonical tensor index

Define

$$\text{tidx}(j, a) = 3j + a, \quad j \in \{0, \dots, 100\}, \quad a \in \{0, 1, 2\}. \quad (14)$$

The inverse is

$$j = \left\lfloor \frac{\text{tidx}}{3} \right\rfloor, \quad a = \text{tidx} \bmod 3. \quad (15)$$

Thus tidx is an exact bijection between $\mathbb{Z}_{101} \times \mathbb{Z}_3$ and $\{0, \dots, 302\}$.

5 Exact monomial step and Gaussian phases

Let

$$D_3 = \text{diag}(1, -2, 1), \quad X_3|a\rangle = |a + 1 \bmod 3\rangle, \quad (16)$$

and define the local twisted step

$$F_3 = X_3 \exp(-i\pi D_3/2). \quad (17)$$

The generalized shift belongs to the standard finite Weyl/Pauli construction [6]. Let $R_{101}|j\rangle = |j + 1 \bmod 101\rangle$ be the declared selector-order cyclic shift. Version 3 defines

$$\boxed{F_{303} = R_{101} \otimes F_3}. \quad (18)$$

At the basis-support level,

$$(j, a) \mapsto (j + 1 \bmod 101, a + 1 \bmod 3). \quad (19)$$

The phase multiplier applied at fibre label a is

$$\exp(-i\pi d_a/2) = i^{g_a}, \quad (d_0, d_1, d_2) = (1, -2, 1), \quad (20)$$

with the exact table

a	d_a	$g_a \bmod 4$	i^{g_a}
0	1	3	$-i$
1	-2	2	-1
2	1	3	$-i$

The aligned phase-symbol array is therefore $(-i, -1, -i)$. The separate four-value lookup table is $(1, i, -1, -i)$. Conflating those arrays would be a contract error.

Lemma 3 (Three-step phase closure). *The accumulated local phase over one fibre cycle is one.*

Proof. The exponent sum is

$$3 + 2 + 3 = 8 \equiv 0 \pmod{4}, \quad (21)$$

so the product is $i^8 = 1$. $\square$

6 Exact order 303

Theorem 4 (Order of the declared monomial step). *The operator F_{303} has exact order 303:*

$$F_{303}^{303} = I_{303}, \quad F_{303}^n \neq I_{303} \quad (1 \leq n < 303). \quad (22)$$

Proof. If the support action returns to its initial address after n steps, then

$$n \equiv 0 \pmod{101}, \quad n \equiv 0 \pmod{3}. \quad (23)$$

Because $\gcd(101, 3) = 1$, the least positive solution is $\text{lcm}(101, 3) = 303$. At 303 steps the fibre cycle occurs 101 times. Each three-step fibre cycle has phase one by the preceding lemma, so the accumulated phase is also one. Therefore the complete monomial operator closes at 303 and at no smaller positive power. $\square$

The implementation additionally exhausts every positive power below 303 and checks 303-step closure from every basis state. Exhaustive finite verification is a regression receipt; the arithmetic proof is the mathematical reason.

7 Canonical event ordering and CRT inverse

Starting at $(0, 0)$, define event $n \in \{0, \dots, 302\}$ by

$$j(n) = n \bmod 101, \quad a(n) = n \bmod 3. \quad (24)$$

Theorem 5 (Event-address bijection). *The map $n \mapsto (j(n), a(n))$ is a bijection from $\mathbb{Z}_{303}$ to $\mathbb{Z}_{101} \times \mathbb{Z}_3$.*

Proof. This is the Chinese remainder isomorphism because 101 and 3 are coprime. An explicit inverse is

$$n = j + 101k, \quad k = 2(a - (j \bmod 3)) \bmod 3, \quad (25)$$

where 2 is the inverse of $101 \equiv 2 \pmod{3}$. $\square$

Each canonical event records:

- sequence index n ;
- tensor index $3j + a$;
- base site index j ;
- external fibre label a ;
- exact transition exponent and Gaussian-unit symbol;
- next sequence index $(n + 1) \bmod 303$; and
- next tensor index.

The 303-entry event table in the accompanying workbook is a complete finite receipt of this bijection.

8 Site registry and preserved root identity

The canonical event document embeds a 101-entry site registry. Each entry contains:

- exact doubled E8 root coordinates;
- preserved v2 basis class: fixed singlet or triality-orbit state;
- fixed index or orbit index where applicable;
- internal triality label and authored low/mid/high display lane where applicable;
- exact selected-graph degree;
- lifted degree $d_j + 2$;
- exact degree-potential numerator $101d_j - 3374$ and denominator 2181; and

- preserved v2 symbolic MIDI note.

The selected basis contains 41 roots from the integer-coordinate family and 60 from the half-integer family. These are exact consequences of the preserved lexicographic selector, not E8 invariants independent of that selector.

9 Cartesian graph lift

Definition 6 (Lifted graph). Let C_3 be the three-vertex cycle. The v3 graph context is

$$\boxed{G_{303} = G_{101} \square C_3.} \quad (26)$$

Its vertices are pairs (j, a) . Horizontal edges copy every base edge at fixed fibre label; vertical edges connect the three fibre labels at fixed site.

Cartesian graph products and their matrix identities are standard [7]. With A_1, L_1 denoting the adjacency and Laplacian of G_{101} , and A_C, L_C those of C_3 ,

$$A_3 = A_1 \otimes I_3 + I_{101} \otimes A_C, \quad (27)$$

$$L_3 = L_1 \otimes I_3 + I_{101} \otimes L_C. \quad (28)$$

Proposition 7 (Exact graph fixtures). *The lifted graph has*

$$|V(G_{303})| = 101 \cdot 3 = 303, \quad (29)$$

$$|E(G_{303})| = 3 \cdot 1687 + 101 \cdot 3 = 5364. \quad (30)$$

For a base vertex of degree d_j , every lifted copy has degree $d_j + 2$. Consequently the degree range is 24–57 and the degree sum is $2 \cdot 5364 = 10728$.

Proof. A Cartesian product contains one copy of every G_{101} edge in each of three fibre layers and one copy of every C_3 edge at each of 101 base vertices. The degree formula for Cartesian products is $\deg_{G \square H}(g, h) = \deg_G(g) + \deg_H(h)$; every C_3 vertex has degree two. $\square$

Since both factors are connected, G_{303} is connected. Its exact degree distribution is:

Degree	24	25	32	34	35	36	42	44	45	46	57
Count	36	18	36	72	36	18	18	36	18	9	6

The data workbook also records the exact finite breadth-first-search receipt $\text{diam}(G_{101}) = 3$ and hence $\text{diam}(G_{303}) = 4$. These diameter values are supplementary exact receipts, not canonical contract identity fields.

9.1 Structural spectral consequence

If λ_i are the Laplacian eigenvalues of G_{101} , then the Laplacian spectrum of G_{303} is the multiset

$$\{\lambda_i, \lambda_i + 3, \lambda_i + 3 : i = 1, \dots, 101\}, \quad (31)$$

because $\text{spec}(L_{C_3}) = \{0, 3, 3\}$. This is an exact symbolic graph-product identity. No floating eigensolver output is used as a selector, contract field, or canonical receipt. Graph signal processing may be applied in separately qualified analyses [8, 9].

10 Canonical event document

The canonical mathematical output is the event document, not a MIDI file and not audio. Its top-level content comprises:

- model and profile identity;
- immutable v2 lineage;
- tensor state-space semantics;
- exact step and phase arrays;
- exact graph context;
- the 101-entry site registry;
- the 303 canonical events; and
- the claim boundary.

The event document is committed by the domain-separated hash

$$\text{d0ab60df11084a230e641d737ec22b2b2ca7d3a3ba5b8759ba669ac4d5254dc0.} \quad (32)$$

The event-array-only receipt is

$$\text{24190612457279577625c0dd2dd4df45bb41c820ee221b31a80f70fde165e3dc.} \quad (33)$$

These hashes establish identity within the stated serialization and domain rules; they do not establish authorship, scientific validity, or physical realization.

11 Receiver artifacts and root output policy

All reference receivers consume the same committed event document. Parameter-mapping discipline requires source fields, target fields, scaling, timing, and receiver assumptions to remain explicit [10].

Artifact	Receiver profile	Exact role
<code>events.json</code>	<code>canonical-json-v1</code>	Canonical core event document and primary event commitment.
<code>events.csv</code>	<code>csv-table-v1</code>	Lossless fixed-column event table suitable for audit and spreadsheet import.
<code>graph.json</code>	<code>cartesian-graph-json-v1</code>	Exact vertices, degrees, and 5,364 labelled edges of $G_{101} \square C_3$.
<code>event-atlas.json</code>	<code>event-atlas-json-v1</code>	Authored integer-grid display coordinates and labels; not a unique geometric embedding.
<code>events.mid</code>	<code>symbolic-midi-channel-v1</code>	External fibre mapped to zero-based MIDI channel and preserved v2 symbolic note mapped to note number.

The root workflow may persist only `.json`, `.csv`, and `.mid` artifacts. It emits no GraphML, SVG, NDJSON, PCM, WAV, or other rendered-audio format. External applications may convert committed JSON to OSC, DMX, SVG, GraphML, haptics, or other receivers; such transforms are separately versioned observation profiles, not canonical root artifacts.

The MIDI profile uses one integer tick per event, velocity 64, and no tempo meta-event. Therefore MIDI timing in seconds and all acoustic realization remain receiver-defined. Low/mid/high and fibre/channel choices are authored symbolic conventions, not intrinsic energy levels or E8-selected acoustics.

12 Canonical serialization and domain-separated hashes

Identity-bearing JSON follows these rules:

1. permitted values are null, booleans, strings, arrays, plain objects, and safe integers;
2. object keys are sorted recursively in lexicographic order;
3. array order is preserved;
4. files use UTF-8, no byte-order mark, and no trailing newline where canonical bytes define identity; and
5. floating-point identity values are rejected.

These rules narrow the implementation profile beyond general JSON syntax [11].

Every semantic receipt uses SHA-256 [12] with domain separation

$$\text{SHA256}(\text{UTF8}(\text{domain}) \parallel 0x00 \parallel \text{payload bytes}). \quad (34)$$

Selected identity receipts are:

Object	SHA-256
Preserved v2 selected basis	97cfd1f087745422fd66d3640c7b86c3209593c4b53741018c08a5e9cdb15f6f
Preserved v2 selected adjacency	29ae0af5b1090c9de30f1efc25789060fb1791eb175d2afcd6888847f7fe6324
Preserved v2 degree-potential numerators	517b9b86e07a1788cab446be93e573d21719af3c1e0eeaedae00b81689c4468
V3 site registry	482ff966b77f02cc28b9d4d0fa4bcfc5e355af97dd35dfefabfce4b28637cb05d
V3 lifted edge list	7c2c820468e3492932ea2d97331114a0c9644d28f9048b864bc81abca84202cf
V3 event array	24190612457279577625c0dd2dd4df45bb41c820ee221b31a80f70fde165e3dc
V3 event document	d0ab60df11084a230e641d737ec22b2b2ca7d3a3ba5b8759ba669ac4d5254dc0
V3 contract payload	7f0330e5c7182fa80acdb09d9bc240c273e78fa9d9ac73f95e58ac9e488d9890
V3 implementation source bundle	2b99e929ba604e44ab9391d4a04ac3314188b13d0c902d603829f13639cc10b6

13 Implementation identity and acyclic provenance

Semantic version 3.0.0 alone is insufficient to distinguish two source revisions that produce different artifacts. The contract therefore records a normalized implementation identity named `etq-303-node-reference-v1`. The source set is:

Source file	Bytes	SHA-256
<code>src/etq-v3-core.mjs</code>	15,443	1ed950cba76551a0f01ca0f5048606d901623e47c689a88460cb8620f6c88c9c
<code>src/etq-v3-canonical.mjs</code>	2,049	fe0af75423c0ac64267a4e4779cdea079c5877bf77bb9caaf62a9b074ed5b2de
<code>src/etq-v3-receivers.mjs</code>	7,373	eb901643c6cc47b30484164e504d5d327a4cfc86e1c185ca1e059b27b5d85eca
<code>src/etq-v3-artifacts.mjs</code>	14,600	3253113d0bf80f72f036910b35025013abc64ca8cce92322b0cf2d36bdd0923d

Source file	Bytes	SHA-256
<code>scripts/build-v3-artifacts.mjs</code>	3,086	b7d9222741ad6029f826745977210042f87c0ed056 1d4ad3c6a9a2987a9734a2

Files are read as UTF-8 text, with CRLF and CR normalized to LF before per-file hashing and ordered source-bundle commitment. The same implementation identity appears in the contract, observation receipt, and manifest lineage.

The provenance chain is deliberately acyclic:

```

preserved v2 fixtures + v3 implementation identity
-> site registry + lifted edges + 303 events
-> canonical event-document commitment
-> deterministic receiver bytes
-> observation receipt
-> manifest core + manifest-core receipt

```

No document claims an ordinary hash of final bytes that already contain that same hash. The documentation-only v3.0.1 release deliberately leaves the v3.0.0 implementation identity and all deterministic receipts unchanged. The receipt architecture is methodologically informed by the separately archived receipt-bound observation protocols [13, 14]; those records are method references, not mathematical dependencies of ETQ-303.

14 Build safety

The build command writes only to a dedicated subdirectory beneath `dist/`. A custom path outside `dist/`, the `dist/` root itself, or an existing non-empty directory is rejected. The builder does not recursively delete a caller-selected path and uses exclusive file creation. These are operational safety properties, not mathematical consequences of E8, but they are part of the reproducible export contract.

The generated bundle places `contract.json` beside `contract.schema.json`; the generated contract therefore uses `./contract.schema.json`. The committed repository fixture uses `../spec/etq-303.v3.schema.json`. The `$schema` location is excluded from the semantic contract payload hash, so a location-correct bundle reference does not change the mathematical contract identity.

15 Breaking change from v2

Version 3.0.0 was a semantic expansion, not a silent rewrite of v2. Version 3.0.1 introduces no further semantic or protocol change.

Topic	ETQ-101 v2.0.0	ETQ-303 v3.0.0
Forward state space	101 root-indexed graph states	$101 \times 3 = 303$ root-indexed tensor states
Ternary structure	33 internal triality/qutrit blocks plus two singlets	Independent external fibre on every one of 101 sites; internal labels remain separate
Canonical observation	101-entry symbolic MIDI code-book	Receiver-neutral 303-entry event document
MIDI	Implemented symbolic code-book	One optional deterministic receiver using channel plus preserved v2 note
Graph context	Selected graph G_{101}	Exact Cartesian lift $G_{101} \square C_3$

Topic	ETQ-101 v2.0.0	ETQ-303 v3.0.0
Order 303	Optional appendix extension	Exact named monomial step with proved order 303
Provenance	Contract and fixture receipts	Contract, implementation identity, event commitment, receiver receipts, manifest
Acoustic output	Excluded	Still excluded

The v1 and v2 contracts remain immutable historical identities. A v3 citation must not be substituted for a v2-specific result, and vice versa.

16 Auxiliary Tanner and spectral modules

The repository retains Tanner-graph spectral tuning tools, grouped eigenspace projectors, non-backtracking operators, and nominal frame-bin calibration experiments. Their graph ratios and numerical receipts may support diagnostics, external receivers, or future conditional tests, but they do not define canonical v3 event identity. In particular:

- no floating eigenvalue or eigenvector enters the v3 contract;
- no frame-bin calibration supplies a root-model clock or hertz scale;
- no Tanner graph replaces the selected E8-root graph;
- no golden-ratio occurrence in a test graph restores a golden modulation profile; and
- no auxiliary numerical result validates a physical E8, QEC, or acoustic system.

17 Reproducibility and verification

A conforming source checkout requires Node.js 20 or later and runs:

```
npm test
npm run verify
npm run build:v3
```

The test and verification chain checks:

1. 240 unique E8 roots with doubled squared norm eight;
2. order-three, orthogonal, root-preserving embedded triality;
3. orbit decomposition $12 + 76(3) = 240$;
4. preserved selector size and v2 basis, adjacency, and degree-potential receipts;
5. tensor-index and inverse bijection on 303 states;
6. CRT event traversal and inverse;
7. phase exponents $(3, 2, 3)$ and aligned symbols $(-i, -1, -i)$;
8. no positive operator power below 303 is the identity;
9. 303-step closure from all 303 basis states;
10. exact graph-product vertices, edges, degree sequence, and connectivity;
11. event-document, site-registry, event-array, and edge-list hashes;
12. allowlisted root artifact extensions only;
13. normalized implementation identity;
14. generated bundle schema portability;
15. unsafe output-path rejection and exclusive writes;
16. acyclic receipt and manifest construction; and
17. continued verification of the preserved v2 and v1 contracts.

18 Zenodo v3.0.1 deposit package

The documentation-only v3.0.1 publication update contains:

- ETQ-303_formal_model_v3.0.1.pdf;
 - ETQ-303_formal_model_v3.0.1.tex;
 - ETQ-303_references.bib;
 - a regenerated UPLOAD_SHA256SUMS.txt; and
 - , if a convenience bundle is used, ETQ-303_Zenodo_Submission_v3.0.1.zip.
- Retain the unchanged data artifacts under their original names:
- ETQ-303_data_table_v3.0.0.xlsx; and
 - ETQ-303_parameter_table_v3.0.0.csv.

They describe the unchanged v3.0.0 protocol identity and deterministic data. The published v3.0.1 record is [10.5281/zenodo.21494678](https://doi.org/10.5281/zenodo.21494678); it was created as a distinct Zenodo version while DOI [10.5281/zenodo.21455181](https://doi.org/10.5281/zenodo.21455181) remains the immutable original v3.0.0 release. The checksum file intentionally does not contain its own checksum, avoiding a circular self-hash.

The spreadsheet contains the 101-entry site registry, 303 events, 5,364 lifted edges, parameter and claim tables, degree distributions, formulas that verify key counts, and the implementation/hash receipts. The CSV is the compact reviewer-facing parameter table.

19 Claim matrix and reviewer-facing limits

Topic	Supported wording	Do not claim
E8 roots	101 preserved selected E8-root labels, each paired with three external fibre states	303 distinct E8 roots
State space	$\mathcal{H}_{303} = \mathcal{H}_{101} \otimes \mathbb{C}^3$	A 303-dimensional representation of the E8 Lie algebra
Dimension terminology	Finite-dimensional algebraic vector spaces; $101 \times 3 = 303$ basis-indexed states	Physical spacetime dimensions or an implied physical ontology
Ternary labels	Internal triality and external fibre are separate fields	The two labels are the same physical degree of freedom
Exact step	Finite monomial action with exponents $(3, 2, 3)$ and symbols $(-i, -1, -i)$	A measured physical evolution
Order 303	Exact order of the declared product step	Periodicity of an arbitrary generator, receiver, or audio renderer
Graph	Exact Cartesian product $G_{101} \square C_3$	The event sequence follows graph edges
Receivers	Deterministic JSON, CSV, and MIDI serializations	Receiver choices are forced by E8
Graph/atlas JSON	Exact graph data and authored integer-grid coordinates	A unique natural embedding of E8 geometry
MIDI	Symbolic channel/note code	Canonical pitch, tempo, loudness, or hertz
Hashes	Identity under stated serialization domains	Proof of authorship, validity, or physical truth
Implementation identity	Normalized hash of named source files	Self-authenticating authorship proof

Topic	Supported wording	Do not claim
Receipt protocol	Acyclic provenance binding	Quantum measurement, collapse, or observer causation
Tanner numerics	Optional auxiliary numerical tools	Canonical v3 identity or exact spectral proof
Audio	Outside the v3 root contract	WAV or PCM as a canonical ETQ-303 artifact

20 Conclusion

ETQ-303 v3.0.1 clarifies the exposition of the bounded, exact, and reproducible finite event protocol first released as v3.0.0. The central result is not “303 E8 roots” or 303 physical dimensions, but a 303-state tensor basis over 101 preserved root labels, equipped with an exact order-303 monomial step and an exact Cartesian graph context. The canonical artifact is the event document; MIDI is one symbolic receiver among several deterministic views.

The formal contribution is the separation of preserved mathematics, exact consequences, authored protocol choices, receiver encodings, numerical auxiliaries, and excluded physical claims. This separation makes the model independently reproducible without converting designed mappings into claims of natural necessity.

A Canonical parameter summary

Parameter or fixture	Canonical value	Origin/status
Model/profile	ETQ-303@3.0.0 / full-qut rit-event-protocol-v1	Named release identity
Documentation version	3.0.1	Terminology and exposition clarification only
Ambient roots	240	Preserved exact E8 realization
V2 orbit decomposition	12 fixed; 76 three-cycles	Exact enumeration
Preserved selector	first 2 fixed; first 33 complete orbits	Authored lexicographic rule
Base dimension	101	Exact after selector
External fibre	3	Authored v3 extension
Total dimension	303	Exact tensor dimension
Tensor index	$3j + a$	Authored canonical ordering; exact bijection
SCL stencil	$(1, -2, 1)$	Preserved exact diagonal
Phase exponents	$(3, 2, 3)$	Exact modulo-four consequence
Phase symbols	$(-i, -1, -i)$	Exact lookup
Support action	$(j, a) \mapsto (j + 1, a + 1)$ modulo $(101, 3)$	Exact finite permutation
Operator order	303	Exact support and phase proof
Event address	$(n \bmod 101, n \bmod 3)$	Exact CRT bijection
Base graph	101 vertices; 1,687 edges; connected	Preserved v2 fixture
Lifted graph	303 vertices; 5,364 edges; connected	Exact Cartesian product
Lifted degree range	24–57	Exact $d_j + 2$ rule
Allowed root outputs	.json, .csv, .mid	Root output policy

Parameter or fixture	Canonical value	Origin/status
Canonical event artifact	<code>events.json</code>	Authored receiver-neutral commitment
Absolute frequency	null / undefined	Explicitly excluded
Rendered audio	disabled in root workflow	Explicitly excluded
Implementation bundle	2b99e929ba604e44ab9391d4a04a c3314188b13d0c902d603829f136 39cc10b6	Normalized source identity

B Required deterministic checks

A conforming independent implementation should check all of the following:

1. exact 240-root E8 generation and doubled norm eight;
2. exact embedded triality and orbit decomposition;
3. preserved 101-label selector and ordering;
4. preserved v2 fixture hashes;
5. 303 tensor addresses and index inverse;
6. 303 CRT event addresses and inverse;
7. aligned phase exponents and symbols;
8. exact order 303 and closure on all basis states;
9. exact site-registry labels and degree-potential numerators;
10. exact 5,364-edge Cartesian lift and degree distribution;
11. connectedness and zero duplicate edges;
12. canonical event document and domain-separated hashes;
13. allowlisted artifact extensions only;
14. normalized source implementation identity;
15. portable generated schema reference;
16. safe output-path behavior; and
17. independent v2 and legacy verification.

C Optional extension boundaries

External receivers may map the committed event document to OSC, DMX, graphics, haptics, robotics, graph-signal arrays, or rendered sound. Each extension must declare its complete mapping, units, timing, numeric ABI, runtime requirements, and receipts. Such a receiver is not part of ETQ-303 v3 canonical identity unless introduced by a new named profile.

An open-system or continuous generator may also be declared separately, but it must not be inferred from the order-303 monomial step. In particular, the exact identity $F_{303}^{303} = I$ does not imply periodicity for an arbitrary Hamiltonian, matrix exponential, Floquet composition, interpolation, or audio renderer.

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